# 5.1 Maximum and Minimum Values; Critical Numbers

Name _____ Period ____ Date _____

AP Calculus 5.1 (30-minute plan) · Student sheet

**I can…**

- tell absolute extrema from local extrema.
- find the critical numbers of a function.
- find the absolute maximum and minimum of a continuous function on a closed interval.

**Vocabulary** · fill in the missing word

**Absolute maximum** — $f(c) \geq f(x)$ for every x in the _____

**Local maximum** — $f(c) \geq f(x)$ for every x in some open interval _____ c

**Critical number** — a number c in the _____ where $f'(c) = 0$ or $f'(c)$ does not exist

**Extreme Value Theorem** — a _____ function on a closed interval [a, b] has an absolute maximum and an absolute minimum

## Part 1 · Notes and examples (fill in each note, then work the example)

**Extreme Value Theorem:** if f is continuous on [a, b], then f has an absolute maximum and an absolute _____ on [a, b].

**Critical numbers:** local extrema can only occur where $f'(c) = 0$ or $f'(c)$ _____.

### Example 1

Find the critical numbers of $f(x) = x^3 - 6x^2 + 9x + 2$.

**Closed interval method:** evaluate f at the critical numbers and at the _____; the largest value is the absolute max, the smallest the absolute min.

### Example 2

Find the absolute extrema of $f(x) = x^3 - 3x$ on $[-3, 3]$.

### Example 3

Find the critical numbers of $f(x) = x^{2/3}$.

## Part 2 · Partner practice

### Partner practice

(AP® Practice) The absolute maximum value of $f(x) = x^3 - 3x^2 + 1$ on $[-1, 3]$ is

(A) 1  (B) 3  (C) −3  (D) 5